



DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF MANAGEMENT & INFO. SCIENCE
SECOND SESSIONAL EXAMINATION

SUBJECT: (23MCA2002) OBJECT ORIENTED PROGRAMMIN WITH JAVA

Examination : MCA Semester – 2
Date : 18/03/2025
Time : 2:00 pm to 3:15 pm

Seat No. :
Day : Tuesday
Max. Marks : 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

- Q.1** **[12]**
- CO1 U (a) Do as directed.**
- CO3 U 1** Which of the following does not allowed duplicate values. **[1]**
 (a) List (b) ArrayDeque (c) Hashset (d) None of these
- CO3 U 2** _____ method of an Object class converts the object into string. *toString()* **[1]**
- CO3 U 3** ArrayList collection is more faster than Array. State True/False **[1]**
- CO3 U 4** Write a syntax for method which is used to update the value of ArrayList elements. *list.set(0, "mang")* **[1]**
- CO3 U 5** Techniques to converts the primitive datatype into wrapper class is called as _____ **[1]**
- CO1 A 6** _____ access modifier allows a class member to be accessed from any class in same package but not from outside of package. **[1]**
 (a) public (b) private (c) protected (d) default
- CO1 A 7** _____ method cannot be override by derived class in inheritance. **[1]**
 (a) final only (b) static only (c) final or static (d) final and static
- CO1 A 8** Which of the following is true about inheritance in Java? **[1]**
 (a) We cannot override private methods
 (b) Protected methods are final.
 (c) Protected members are accessible within a package and inherited classes outside the package
 (d) A class extends two or more classes.
- CO1 A 9** Static data member of a class can access via its class name. State True/False **[1]**
- CO1 A 10** Lets consider A and E are classes. B and D are interfaces and C is an abstract class. Which of the following is true. **[1]**
 (a) Class F implements B, C {} (b) Class F extends A,E {}
 (c) Class F impliments B,D {} (d) None of these
- CO1 A 11** Find the error if any else write the output **[1]**
- ```

class Derived
{
 protected static void getDetails()
 {
 System.out.println("Derived class");
 }
}

```

```

public class Test extends Derived
{
 protected static void getDetails()
 {
 System.out.println("Test class");
 }
 public static void main(String[] args)
 {
 Derived obj = new Derived();
 obj.getDetails();
 }
}

```
- CO1 A 12** Find the output **[1]**
- ```

class A{
    A() {System.out.println("A");}
    void print()
    {System.out.println("print A");}
}
class B extends A
{
    B() {
        System.out.println("B");
    }
    void print(){
        System.out.println("Inside B");
    }
}

```

```

class Test {
    public static void main(String[] args) {
        A b = new B();
        b.print();
    }
}

```

- Q.2** Attempt *Any Two* from the following questions. [12]
- CO1 A** (a) Illustrate what is Object class and its methods used for. Create the class Car having property modelName,color,price. Create constructor to set the values. Override the method toString() method of Object class. Override equals() method of object class which check if two car objects have same price then it is equal. [6]
- CO1 A** (b) Create abstract class Match having abstract method void cal_performance() [6]
 Create a class Cricket which inherits Match. It has property noofplayer, totalscore. Create constructor to initialize the values. Implement abstract method which calculates average score and display it. (totalscore/noofplayer)
 Create a class Football which inherits Match class. Class has properties noofgoals and timeinminutes. Create constructor to initialize the values. Implement abstract method which will calculate performance by calculating percentage (noofgoals/time)*100) and display it.
 Create a class Chess which inherits Match class. Class has properties oldrating, ActualScore. Create constructor and initialize the value. Implements abstract method and calculate performace of player (oldrating + ActualScore * 1.2) display it.
 Write java code which create object of Cricket,Football and Chess and display performance of each object.
- CO1 A** (c) Create a class Operation, Properties: int[] ivalue, String[] svalue. Create [6]
 constructor to create an array of ivalue and svalue with size 10.
 Class contains following Methods
 void add() : which calculate sum of all elements ivalue and display result.
 void add(String s): it concate all element of svalue and String s and display result.
 boolean search(int i): it returns the true if the value is present into ivalue array.
 boolean search(String s) : it returns the true if String s is present in String array svalue.

Q.3 Do as Directed

- CO3 N** (a) Create class Item having properties name, price. Create necessary constructor to initialize the values. Create getPrice() method which will return the price of an Item. Create a five objects of Item class and add it into the ArrayList Collection. also perform following operations on list [8]
 1. Remove the element from list
 2. Display all elements of the list
 3. Display message "present" if collection have item's price 500. Ow print message "not present"
 Note: use appropriate datatype and return type whenever it is required.
- CO3 A** (b) Explain relative and static imports, explain java packages with suitable example as well steps to compile and run it. [4]

OR

Q-3 Do as Directed

- CO3 N** (a) Create a class Merchant having property name ,city and have constructor to initialize values. class have a method called void printDetail() which print the merchant detail. Create class Item inherited from Merchant class. Item class have property itemname, price. Create constructor to set the values. Item class have method getPrice() that returns the price of an item. Void Display() method print the details of a Item. [8]
 Create class Shipment inherited from Item class. Shipment class contains properties orderno, qtyordered. Create constructor to set values. Method float cal_total() calculate total cost of an order (itemprice*qtyordered). Void Display() Method display all the details including merchant ,item and shipment.
 Write a java code which create an object of shipment class and display all the detail.
- CO3 A** (b) Illustrate the upcasting and downcasting of object. [4]



DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF MANAGEMENT AND INFORMATION SCIENCE
SECOND SESSIONAL

SUBJECT: (23MCA2001) SOFTWARE ENGINEERING AND TESTING

Examination : MCA Semester II
Date : 17-03-2025
Time : 2.00 TO 3.15 P.M.

Seat No. :
Day : Monday
Max. Marks : 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

- Q.1 Do as directed. [12]**
- CO1 U** (a) Consider the following partial requirement for a software: "A web browser should be used as the front end." It can be considered to be which one of the following types of requirements: [1]
(1) Functional requirement (2) Non-functional requirement (3) Constraint
(4) External interface
- CO1 U** (b) Consider the following partial description of the IIT security software. "A camera at the main gate should detect an incoming vehicle, and the gate should be opened for registered vehicles." Which one of the following types of requirements is this? [1]
(1) Functional requirement (2) Non-functional requirement (3) Design requirement (4) Design constraint
- CO1 U** (c) State true or false with justification [Without justification would lead to zero marks] [6]
f 1) A formal specification can be ambiguous.
f 2) The system test plan cannot be prepared immediately after the completion of the requirements specification phase.
T 3) The essence of any good function-oriented design technique is to map the functions performing similar activities into a module.
f 4) The extent to which two modules depend on each other determines the cohesion of the two modules.
- CO1 U** (d) Multiplicity for an association _____ [1]
(i) association is the number of instances with a single instance
(ii) association is the number of instances with a number instance
(iii) All of the mentioned
(iv) None of the mentioned
- CO1 U** (e) Which term are combined Interaction Diagram? [1]
(i) Sequence + Collaboration (ii) Activity + State Chart
(iii) Deployment + Collaboration (iv) None of the mentioned
- CO1 U** (f) An attribute is a data item held by which of the following? [1]
(i) Class (ii) Object (iii) All of the mentioned (iv) None of the mentioned
- CO1 U** (g) The UML supports event-based modeling using _____ diagrams. [1]
(i) Deployment (ii) Collaboration (iii) State chart (iv) All of the mentioned.
- Q.2 Attempt Any TWO from the following questions. [12]**
- CO2 C** (a) Enumerate the different types of cohesion that a module in a design might exhibit. Give examples of each. [6]
- CO2 C** (b) Point out three important differences between the function oriented and the object-oriented approaches to software design. Verify our answer through suitable examples. [6]
- CO2 C** (c) Draw the class diagram for library management system [6]

Q.3 Attempt the following questions.

[12]

- CO3 A (a) In a MIS application following task is mention below [8]
1) Requirement specification 2) Design 3) Code 4) Test 5) Document
6) Database part 7) Graphical user interface part.
Draw the work break down structure of MIS application.

The duration for the task are mentioned below :

Specification (15 days), Integrate and test (120 days), Code GUI (45 days)
Design GUI (30 days), Code database (105 days), Write user manual (60 days)
and Design database (45 days).

Prepare the network diagram and find the critical path and project duration.

- CO2 C (a) From the information given below draw a single diagram representing object association relation [4]
• A teacher teaches 1 to 3 courses
• Each course is taught by only one teacher
• A course can have 10 to 300 students.

OR

- CO3 A (a) 1) For the following code estimate the Halstead's length and volume measures [8]

```
main() {  
    int a,b,c,avg;  
    scanf("%d %d %d",&a,&b,&c);  
    avg=(a+b+c)/3;  
    printf("avg= %d",avg); }
```

- 2) Assume that the size of an organic type software product has been estimated to be 32,000 lines of source code. Assume that the average salary of a software developer is `15,000 per month. Determine the effort required to develop the software product, the nominal development time, and the cost to develop the product.

- CO2 C (b) Prepare the CRC card of a bank customer. [4]

Blooms Taxonomy levels : R-Remembering, U- Understanding, A-Applying, N-Analyzing, E- Evaluating, C- Creating

2) → The system test plan can be prepared after the
false) requirements Phase because testing is planned
against requirement.

3 → True - Function-oriented design groups similar functions
into modules to promote modularity & Cohesion.

4 → False → The degree of dependence bet. modules determines
Coupling, not Cohesion



DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF Management and Information Science
SECOND SESSIONAL
SUBJECT: (MCA-23MCA2004) Python Programming

Examination : MCA Semester **2**
Date : 20.03.2025
Time : 02.00 pm – 03.15 pm

Seat No. :
Day : Thursday
Max. Marks : 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.
5. Write neat source code with NECESSARY indentation and format.
6. Attempt questions in sequence.

Q.1 Do as directed.

[12]

- CO3 A** (a) (i) Implement a lambda function that accepts two numeric parameters, and returns: the largest and the mean of those two parameters. [4]
- (ii) Implement a lambda function that accepts a string parameter and returns: the parameter in upper case and the parameter in lower case.
- CO3 A** (b) (i) Compute factorial of a number using lambda. [8]
- (ii) Compute factorial of a number using reduce.
- (iii) Filter numbers divisible by 5 from a Python list.
- (iv) Given a Python list of marks of 10 students. Using lambda, give additional 3 marks to all students.

Q.2 A Attempt *Any TWO* from the following questions. [12]

- CO2 A** (a) Write a Python program for the following requirements:
- Create a class Number with a data member num to store a number.
 - In the Number class,
 - o create appropriate constructor for initializing the num data member.
 - o create a method calcFactorial() that computes and returns factorial of the data member num.
 - o create a method isPalindrome() that should return True if the num contains a palindrome number, False otherwise.
 - Demonstrate usage of the Number class.
 - Explain your code.
- CO2 A** (b) Write a Python program to create a class representing a Bank Account. Include necessary attributes, constructor and methods for managing customer accounts including deposit-transaction and withdraw-transaction. Take care of sufficient balance availability during the withdraw-transaction.
- CO2 A** (c) (i) How to establish Python-MySQLDB connection?
- (ii) How to access Base class's members in Derived class body?
- (iii) How to use sorted() function?

Q.3 Attempt *Any ONE* from the following questions. [12]

- CO3 A** It is required to implement an Object-Oriented program to store and retrieve Student details into/from a MySQL database table.
- Write a Python program to create a class Person to represent a person, with data members- name & age; and a method- setPerson().
 - From the Person class, derive a class Student with one more data member: semester. In the Student class, implement three instance methods: setStudent(), showStudent() and save(). The setStudent() and showStudent() methods respectively set and show values of the data members name, age and semester.
 - The save() method should save respective Student object's properties in an existing database - by inserting a new row (name, age and semester) in a MySQL table 'tblStudent' in a database

- 'dbCollege' available on localhost. Assume MySQL credentials.
- Also implement a static method `showAllStudents()` in the `Student` class. It fetches all rows from the `tblStudent`, and shows all rows in tabular format on screen.
- Show usage of the various methods available in the `Student` class.

OR

CO3 A

A fruit shopkeeper wants to maintain fruits data in an existing table `tblFruits` in an existing MySQL database `dbFruits`. Help him by providing following functionalities in Python:

- Create a database `dbFruits` in a MySQL server hosted at an IP address 10.10.10.10; assume necessary credentials.
- In the database `dbFruits`, create a table `tblFruits` with columns: `fruitID`, `fruitName`, `ratePerPiece` and `piecesAvailable`. The columns `fruitID`, `ratePerPiece` and `piecesAvailable` store numeric data. The column `fruitName` stores string data.
- Create a stored procedure `spInsertFruit()` to insert a new fruit row in the `tblFruits`. The stored procedure should accept necessary parameters.
- Demonstrate usage of `spInsertFruit()` to insert a new row.
- Create a stored procedure `spFindFruitByName()` to search a fruit by name. The `spFindFruitByName()` should
 - o receive a parameter `paramFruitName`, and
 - o fetch relevant fruit records containing `paramFruitName` anywhere in the fruit name in the `tblFruits` table,
 - o return such fruit records to Python program.
- Demonstrate usage of `spFindFruitByName()`.
 - o Show returned results on screen.
 - o Show "No such fruit available" in case no results are found.
- Write suitable Python code to increase all fruits' price by Rs. 2 per piece in the `tblFruits` table.



DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF MANAGEMENT AND INFORMATION SCIENCES
SECOND SESSIONAL
SUBJECT: (23MCA2003) DATA STRUCTURES

Examination : MCA Semester - II
Date : 19-03-2025
Time : 02:00 PM to 03:15 PM

Seat No. : _____
Day : Wednesday
Max. Marks : 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

Q.1 Do as directed.

[12]

CO3 U (a) Answer the Following

- (i) Five items HARRY, ALBUS, RON, NEWT and SIRIUS are pushed onto a stack one after another starting from ^{Planay} 1. The stack is popped four times and the popped elements are inserted into a queue. The two elements are deleted from the queue and pushed back onto the stack. Again one element popped from the stack. The popped item will be _____.

(A) HARRY (B) ALBUS (C) RON (D) NEWT (E) SIRIUS

- (ii) Suppose, a circular queue of capacity $(n-1)$ elements is implemented with an array of n elements. Assume that the insertion and deletion operations are carried out using REAR and FRONT as array index variables respectively. Initially, REAR=FRONT=0. The conditions to detect queue full and queue empty are _____. [mod is Modulus]

- ✓ (A) full: $(\text{REAR} + 1) \bmod n = \text{FRONT}$ empty: REAR = FRONT
(B) full: $(\text{REAR} + 1) \bmod n = \text{FRONT}$ empty: $(\text{FRONT} + 1) \bmod n = \text{REAR}$
(C) full: REAR = FRONT empty: $(\text{REAR} + 1) \bmod n = \text{FRONT}$
(D) full: $(\text{FRONT} + 1) \bmod n = \text{REAR}$ empty: REAR = FRONT

- (iii) Five people HARRY, ALBUS, RON, NEWT and SIRIUS are standing in a queue. RON is standing between HARRY and SIRIUS. HARRY is just behind ALBUS and ALBUS is second in the queue. Who is second last in the queue?

(A) HARRY (B) ALBUS (C) RON (D) NEWT (E) SIRIUS

- (iv) Typical time requirement for inserting a element at start of linked list with head pointer is _____

- ✓ (A) $O(1)$ (B) $O(n)$ (C) $O(\log n)$ (D) $O(n^2)$

- (v) The following C function takes a singly-linked list of integers as a parameter. The list is represented as a pointer to a structure. What will be the contents of the list after the function completes execution?

//Input List 1001->1002->1003->1004->1005->1006->1007

```
void solve(struct node* list){  
    struct node *p,q;  
    int temp;  
    if (! List || ! list->next) return;  
    p=list;  
    q=list->next;  
    while(q){  
        temp=p->value; p->value=q->value;  
        q->value=temp;p=q->next;  
        q=p?p->next:0;  
    }  
}
```

Handwritten notes: 1002 → 1001 → 1004 → 1003 → 1006 → 1005 → 1007

- (vi) What will be the output of the following code snippets for the list?

[02]

//Input List 250->260->270->280->290->30->70

```
void solve(struct node* start){
```

```

if(start == NULL)
    return;
printf("%d ", start->data);
if(start->next != NULL )
    solve(start->next->next);
printf("%d ", start->data);

```

} 250270290 290270250

- CO3 U (b) Discuss algorithm and Write code to find the middle node of the linked list. Your solution should be able to find the node in a single scan. Assume that you have a Singly Linked List with a head pointer. [04]

```
struct ListNode * locateMidNode(struct ListNode *head)
```

- Q.2 Attempt *Any TWO* from the following questions. [12]

- CO5 C (a) Write a C program to implement a dynamic stack using a linked list. Also write a function to reverse a string using the same stack. [06]

- CO5 C (b) Write a C Program to implement a dynamic circular queue of size N. [06]

- CO5 C (c) Discuss algorithm and Write a method to check whether the given linked list is NULL-terminated or If there is a cycle. If the cycle exists then find the length of the loop. [06]

```
int checkLoop(struct ListNode *head)
```

- Q.3 Attempt the following questions. [12]

- CO1 C (a) Write a program to implement a singly linked list with the following operations. The Node of linked list will hold data for Products [id,name,price,weight]. [06]

(i) Insert at End (ii) Delete at End (iii) Sort List based on price

- CO1 C (b) Write a program to implement an output restricted Deque of size N. [06]

OR

- Q.3 Attempt the following questions. [12]

- CO1 C (a) Create a simple queue for purchasing movie tickets. Write a C program to create and then process the below provided queue for purchasing movie tickets. [06]

```

typedef struct{
    char name[25];
    int tickets_required;
    bool tickets_acquired;
}Person;

```

```

typedef struct{
    Person* pQueue;
    int front, rear;
}MovieQueue;

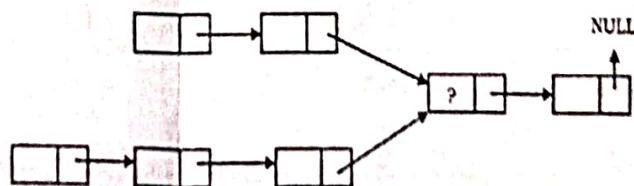
```

Total Available Tickets: 20

Persons in Queue: [HARRY - 5, ALBUS - 10, RON - 2, NEWT - 1, SIRIUS - 5]

- CO1 C (b) Write an algorithm for finding the merging point [merging node] in the linked list. [Use Arrays/ Stack in your solution] [06]

Suppose there are two singly linked lists both of which intersect at some point and become a single linked list. The head or start pointers of both the lists are known, but the intersecting node is not known. Also, the number of nodes in each of the lists before they intersect is unknown and may be different in each list. List1 may have n nodes before it reaches the intersection point, and List2 might have m nodes before it reaches the intersection point where m and n may be $m = n, m < n$ or $m > n$.





DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF MANAGEMENT AND INFORMATION SCIENCES
SECOND SESSIONAL EXAMINATION
SUBJECT: 23MCA-2102 System Analysis, Design and Management

Examination	: MCA-Semester - II	Seat No.	:
Date	: 21/03/2025	Day	: Friday
Time	: 2:00 – 03:15 pm	Max. Marks	: 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
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4. Draw neat sketches wherever necessary.

Q.1 Do as directed.

[12]

CO5 U 1 i) A pseudo code is _____ [4]

- a) A random number b) a flow chart ☒ c) a structured English to communicate logic of the program d) None of these

ii) What of the following is true about Cost-Benefit analysis?

- a) Evaluates the tangible and non-tangible factors
b) Compares cost with benefits of introducing a computer-based system
☒ c) Estimates the hardware and software cost
d) All of the above

iii) Changes made periodically to the system is called _____

- a) Analysis b) Design ☒ c) Maintenance d) Debugging

iv) Which of the following is not part of DFD?

- ☒ a) Disk storage b) Arrow c) Data storage d) bubble circle

2 Discuss the steps to predict overall cost of the project. [3]

3 Differentiate: HIPO and Flow chart. [2]

4 What are the three tools used for software design and documentation? [3]

Q.2 Attempt *Any TWO* from the following questions. [12]

CO4 U (a) Discuss different criteria used for Hardware and Software selection. Also discuss vendor selection process. [6]

(b) What is a HIPO chart? Why is it considered as top-down design approach? Discuss the components of HIPO chart taking same example. [6]

(c) What is file conversion? Discuss types of file conversions with example. [6]

Q.3 Attempt the following questions. [12]

CO6 A (a) Consider the development of an Employee Management System for an organization. The system is aimed at automating the complete employee details including maintenance of employee attendance records, leave details, and printing monthly slips. This system is also expected to sum up each employee's work hours, time off, shift pays that help to generate slip of employees. Draw the context diagram and DFD for developing this system. Explain why you chose certain elements as processes versus sources/sinks. [8]

CO7 U (b) What are the roles and responsibilities of a project manager? Discuss the common interpersonal skills required in a project manager. [4]

OR

CO6 A (a) Estimate and explain benefits of an effective budget chart in system development. What are the disadvantages of not preparing budget chart? [6]

CO7 U (b) Discuss program management, portfolio management and project management. How the three are interrelated? [6]